Kellin N. Rumsey

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Los Alamos, NM 87545

Education

PhD — **Statistics | University of New Mexico** — **Albuquerque, NM**
Cumulative GPA: 4.13 / 4.00 — *2015–2020*
Dissertation: Methods of Uncertainty Quantification for Physical Parameters (Advised by Gabriel Huerta & Lauren Hund)

MS — **Computer Science | University of New Mexico** — **Albuquerque, NM**
GPA: 4.08 / 4.00 (CS courses only) — *2015–2020*

BS — **Mathematics | University of Arizona** — **Tucson, AZ**
Focus in Probability and Statistics, — *2013–2015*
Minors in Computer Science and Chemical Engineering

Research Experience

Los Alamos National Laboratory — **Los Alamos, NM**
Scientist, Statistical Sciences — *2022–Present*

Los Alamos National Laboratory — **Los Alamos, NM**
Postdoctoral Researcher, Statistical Sciences — *2020–2022*

Sandia National Laboratories — **Albuquerque, NM**
Year Round Intern, Statistical Sciences — *2016–2020*

Centers for Research Excellence in Science and Technology — **Albuquerque, NM**
Statistical Consultant — *2017–2020*

UofA Microgravity Research Team — **Tucson, AZ**
Data Analyst — *2013–2014*

Professional Experience

University of New Mexico — **Albuquerque, NM**
Teaching Assistant — *2015–2020*

Cota Enterprises — **Topka, KS**
Quality Assurance Programmer — *2014, 2015*

Refrac Systems — **Chandler, AZ**
Materials Science Engineering Intern — *2013*

(Last updated: October, 2025)

Awards

2025 ***Statistics in Physical Engineering Sciences Award –*** Awarded for innovative use of statistics to solve a high-impact problem in the physical and engineering sciences.

2020 ***NFL Big Data Bowl, Finalist –*** Finalist (Top 3), Collegiate Section.

2018 ***Susan D. Roberts Outstanding Teaching Assistant Award –*** Awarded annually to three graduate students university-wide for excellence in teaching. (Also nominated in 2017.)

2018 ***Ji-Hyun Lee Award –*** Awarded for best student presentation at the annual meeting of the Albuquerque Chapter of the ASA.

2017 ***Best Use of Outside Data –*** ASA DataFest Competition at UNM

2016 ***Graduate Student of the Year Award –*** Awarded by the Department of Mathematics and Statistics, UNM

2013 ***NJCAA All-Academic Honors –*** Recognized for academic achievement while competing in intercollegiate tennis at Mesa Community College.

Publications google scholar

Publications (Statistical Research)

[12] Rumsey, K., Francom, D., Gibson, G. C., Tucker, J. D., & Huerta, G., 2026. Bayesian Adaptive Polynomial Chaos Expansions. *Stat,* In press.

[11] Hutchings, G., Rumsey, K.N., Bingham, D. and Huerta, G., 2025. Enhancing approximate modular Bayesian inference by emulating the conditional posterior. *Computational Statistics & Data Analysis*, *212*, p.108235.

[10] Rumsey, K.N., Hardy, Z.K., Ahrens, C. and Vander Wiel, S., 2025. Co-active subspace methods for the joint analysis of adjacent computer models. *Technometrics*, *67*(1), pp.133-146.

[9] Rumsey, K., Francom, D., Vander Wiel, S., 2024. Discovering Active Subspaces for High-Dimensional Computer Models. *Journal of Computational and Graphical Statistics*, pp.1-13.

[8] Rumsey, K., Francom, D., Shen, A., 2024. Generalized Bayesian MARS: Tools for Stochastic Computer Model Emulation. *SIAM/ASA Journal on Uncertainty Quantification,* 12(2), pp.646-666.

[7] Collins, G., Francom, D., Rumsey, K., 2023. Bayesian Projection Pursuit Regression. *Statistics and Computing*, 34(1), 29.

[6] Rumsey, K.N., Huerta, G. and Derek Tucker, J., 2023. A localized ensemble of approximate Gaussian processes for fast sequential emulation. *Stat*, *12*(1), p.e576.

[5] Rumsey, K., Grosskopf, M., Lawrence, E., Biswas, A. and Urban, N., 2022, October. A hierarchical sparse gaussian process for in situ inference in expensive physics simulations. In *Applications of machine learning 2022* (Vol. 12227, pp. 126-138). SPIE.

[4] Grosskopf, M., Lawrence, E., Biswas, A., Tang, L., Rumsey, K., Van Roekel, L. and Urban, N., 2021. In-situ spatial inference on climate simulations with sparse gaussian processes. In *ISAV'21: In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization* (pp. 31-36).

[3] Rumsey, K.N. and Huerta, G., 2021. Fast matrix algebra for Bayesian model calibration. *Journal of Statistical Computation and Simulation*, *91*(7), pp.1331-1341

[2] Rumsey, K., Huerta, G., Brown, J. and Hund, L., 2020. Dealing with measurement uncertainties as nuisance parameters in Bayesian model calibration. *SIAM/ASA Journal on Uncertainty Quantification*, *8*(4), pp.1287-1309.

[1] Hund, L., Schroeder, B., Rumsey, K. and Huerta, G., 2018. Distinguishing between model-and data-driven inferences for high reliability statistical predictions. *Reliability Engineering & System Safety*, *180*, pp.201-210.

Publications (Statistical Consulting)

[9] Coffman, C.N., Carroll-Portillo, A., Alcock, J., Singh, S.B., Rumsey, K., Braun, C.A., Xue, B. and Lin, H.C., 2024. Magnesium oxide reduces anxiety-like behavior in mice by inhibiting sulfate-reducing bacteria. *Microorganisms*, *12*(7), p.1429.

[8] Scruggs, C.E., Lozoya, S., Rumsey, K.N., Bronson, K. and Chavez, P., 2024. Voluntary public campaigns to benefit the environment: Assessing the effectiveness of the There Is No Poop Fairy campaign. *Journal of Planning Education and Research*, *44*(2), pp.808-821.

[7] Birg, A., Coffman, C., Rumsey, K., Lin, H. and Kanagy, N., 2024. Bismuth subsalicylate reserves high fat diet-induced increase in portal venous flow.

[6] Lin, J.Y., Wang, J., Mishra, K., Osmani, S., Rumsey, K., Singh, P.P. and Barbosa, N.S., 2024. Non–melanoma skin cancer in solid organ transplant recipients with skin of color. *Journal of the American Academy of Dermatology*, *90*(1), pp.159-160.

[5] Scruggs, C.E., Heyne, C.M. and Rumsey, K.N., 2023. Understanding questions and concerns about potable water reuse: An analysis of survey write-in responses. *AWWA water science*, *5*(2), p.e1333.

[4] Carroll-Portillo, A., Rumsey, K.N., Braun, C.A., Lin, D.M., Coffman, C.N., Alcock, J.A., Singh, S.B. and Lin, H.C., 2023. Mucin and agitation shape predation of Escherichia coli by lytic coliphage. *Microorganisms*, *11*(2), p.508.

[3] Distler, L.N., Scruggs, C.E., Cruz, M.P. and Rumsey, K.N., 2021. Public Engagement on Water Reuse Beyond Community Surveys. *Journal: American Water Works Association*, *113*(4).

[2] Lin, J., Rumsey, K., Stepenaskie, S. and Durkin, J.R., 2021. Dermatologists are more likely than oncologists to prescribe skin-directed therapies for early-stage cutaneous T-cell lymphoma: a retrospective review. *Dermatology online journal*, *27*(10).

[1] Distler, L.N., Scruggs, C.E. and Rumsey, K.N., 2020. Arid inland community survey on water knowledge, trust, and potable reuse. II: Predictive modeling. *Journal of Water Resources Planning and Management*, *146*(7), p.04020046.

Technical Reports

[1] Hund, L., Schroeder, B.B., Rumsey, K. and Murchison, N., 2017. Robust approaches to quantification of margin and uncertainty for sparse data *(No. SAND2017--9960). Sandia National Laboratories (SNL-NM),* Albuquerque, NM (United States).

Under Review, In Progress, or Miscellaneous

- Rumsey, K.N., Gibson, G.C., Francom, D. and Morris, R., 2025. All Emulators are Wrong, Many are Useful, and Some are More Useful Than Others: A Reproducible Comparison of Computer Model Surrogates. *arXiv preprint arXiv:2512.09060*.

- Grosskopf, M., Rumsey, K., Biswas, A. and Lawrence, E., 2025. A Partitioned Sparse Variational Gaussian Process for Fast, Distributed Spatial Modeling. *arXiv preprint arXiv:2507.16771*.

- Hutchings, G., Bingham, D., Rumsey, K. and Lawrence, E., 2024. Fast Emulation, Modular Calibration, and Active Learning for Simulators with Functional Response. *arXiv preprint arXiv:2405.16298*.

- Wright, C. and Rumsey, K., 2020. The strengths, weaknesses and promise of differential privacy as a privacy-protection framework. *University of New Mexico. URL: https://math. unm. edu/~knrumsey/pdfs/projects/DifferentialPrivacy. pdf*.

Software

duqling: An R package for reproducible UQ research (Author)

coactivity: An R package for active and co-active subspaces (Author)

GBASS: An R package for generalized Bayesian MARS emulation (Author)

khaos: An R package for Bayesian polynomial chaos expansions (Author)

conforest: An R package for conformal random forests (Author)

leapgp: An R package for fast sequential emulation (Author, CRAN)

spareGParts: An R package implementing various approximate GP algorithms (Author/Host)

PRISM: A toolkit, written mostly in Julia, for in-situ inference in HPC settings (Contributor)

RQMU: A R package for widespread use at Sandia National Laboratory (Author, private)

Presentations

Invited Presentations

[16] Enthusiastic Collaboration and Mentoring. *Gerald J. Hahn Achievement Award* Plenary Talk. Fall Technical Conference, Houston, TX. 2025. (Presented by Scott Vander Wiel)

[15] `duqling`: A Framework for Reproducible Benchmarking of Emulators. Talk. Spring Research Conference, New York City, NY. 2025.

[14] Generalized BMARS, Co-Active Subspaces, and Polynomial Chaos. Seminar. TTO, Statistical Sciences Department, Los Alamos National Laboratory. 2025.

[13] Co-Active Subspace Method for the Joint Analysis of Adjacent Computer Models. Talk. INFORMS Annual Meeting, Seattle, WA. 2024.

[12] Dimension Reduction and Sensitivity Analysis: New Subspace Methods for Complex Computer Models. Talk. Weapons Systems Capability Review (Early Career Session), Los Alamos National Laboratory. 2024.

[11] Discovering Active Subspaces for High Dimensional Computer Models. Talk. INFORMS Annual Meeting, Phoenix, AZ. 2023.

[10] Distributed Sparse Variational GPs: Independent Beats Hierarchical… but Why? Seminar TTO, Statistical Sciences Department, Los Alamos National Laboratory. 2022.

[9] A Hierarchical Sparse Guassian Process for In Situ Inference in Expensive Physics Simulations. Talk. SPIE, San Diego, CA. 2022.

[8] Active Subspace Methods. Talk. Los Alamos - Arizona Days. Los Alamos, NM. 2022.

[7] Generalized Bayesian MARS: Robust Regression, Quantile Regression, and More. Seminar. TTO, Statistical Sciences Department, Los Alamos National Laboratory. 2021.

[6] Quantifying Leverage at the Point of Attack. Talk (with Brandon DeFlon). NFL Big Data Bowl, Indianapolis, IN. 2020.

[5] A Dynamic Bayesian Approach to Influenza Modeling. Talk. UNM SIAM Chapter Meeting. Albuquerque, NM. 2019.

[4] Probability of Prior Coherency and Moment Penalization. Seminar. MARTIANS Symposium, Sandia National Laboratory. 2019.

[3] Dealing with Nuisance Parameters in Bayesian Model Calibration. Talk (with Gabriel Huerta). UARK Spring Lecture Series, Fayetteville, AR. 2019.

[2] Classification and Outlier Detection with Functional Extremal Depth. Seminar. Time Series Data Mining Research Group, University of New Mexico. 2019.

[1] Using Zipf's Law to Control for Voluntary Response Bias in Twitter Data. Seminar. Tractor Talks, Department of Mathematics and Statistics, University of New Mexico. 2016.

Contributed Presentations

[13] An Extensive and Reproducible Comparison of Computer Model Emulators. Poster. Joint Statistical Meetings (JSM), Nashville, TN. 2025.

[12] Co-Active Subspaces for the Joint Analysis of Adjacent Computer Models. Talk. Sensitivity Analysis of Model Output (SAMO), Grenoble, France. 2025.

[11] Bayesian MARS: Emulation, Sensitivity, and Dimension Reduction. Talk. Joint Statistical Meeting (JSM), Toronto, Canada. 2024.

[10] A Partitioned Sparse Variational Gaussian Process for Fast, Distributed Spatial Modeling. Talk. SIAM/ASA Conference on Uncertainty Quantification, Trieste, Italy. 2024.

[9] Fast Distributed Spatial Modeling for Large Scale Climate Models. Talk. Joint Statistical Meetings (JSM), Washington, DC. 2023.

[8] Discovering Active Subspaces for High-Dimensional Computer Models. Poster. Conference on Data Analysis (CoDA). Santa Fe, NM. 2023.

[7] A Hierarchical Sparse Gaussian Process for In Situ Inference in Expensive Physics Simulations. Poster. Climate Model Evaluation and Uncertainty (IMSI Workshop). Chicago, IL. 2022.

[6] Generalized Bayesian MARS. Talk (speed). Joint Statistical Meetings (JSM), Virtual, 2021.

[5] Methods of Uncertainty Quantification for Physical Parameters. Poster. UNM Shared Knowledge Conference, Albuquerque, NM. 2019.

[4] Dealing with Measurement Errors as Nuisance Parameters in Bayesian Model Calibration. Talk. Uncertainty Quantification Mini Workshop, Austin, TX. 2018.

[3] Ridge Regression, LASSO, and Beyond (for Bayesians). Seminar. MARTIANS Symposium, Sandia National Laboratory. 2018.

[2] Dealing with Nuisance Parameters in BMC: Regularization vs Modularization. Poster. Conference on Data Analysis (CoDA). Santa Fe, NM. 2018.

[1] Dealing with Nuisance Parameters in BMC: Regularization vs Modularization. Talk. Albuquerque Chapter ASA Meeting. Santa Fe, NM. 2017.

Mentoring

Student Interns

- Reid Morris, Northeastern, undergraduate student, 2025 (official, joint with Devin Francom)
- Grant Hutchings, Simon Fraser, PhD student, 2023-2025 (joint with James Gattiker and Derek Bingham)
- Kendric Hood, undergraduate student, 2024 (joint with Nathan Debardeleban)
- Jared DiDomenico, UNM, PhD student, 2023 (official, joint with Devin Francom)
- Andy Cox, undergraduate student, 2023 (joint with Ayan Biswas)
- Gavin Collins, Ohio State, PhD student, 2021—2022 (official, joint with Devin Francom)
- Andy Shen, UCLA, undergraduate student, 2020 (joint with Devin Francom)

Professional Service

Professional Societies

2018-23 *Secretary* — Albuquerque Chapter, American Statistical Association

2023—25 *Co-President* — UNM Department of Mathematics and Statistics, Graduate and Professional Student Association

Organizing Committees

2017 *Organizer* — UNM Tractor Talks

2026 *Organizing committee* — Quality and Productivity Research Conference

Other Service

2023, 2025 Session Chair – Joint Statistical Meetings Conference

2024, 2025 Committee Member – Laboratory Directed Research and Development

2025 Committee Member – L2 Milestone Review (LLNL)

2019 Science Fair Judge – New Mexico Tech (State Championships)

2019 Top contributor – Cross Validated Stack Exchange (Ranked 51 / 6618)

2018 Panel Member – Southwestern Undergraduate Mathematics Research

Conference

Peer review referee:

- *Bayesian Analysis (5)*
- *Technometrics (3)*
- *SIAM/ASA Journal on Uncertainty Quantification (3)*
- The R Journal
- Statistical Analysis and Data Mining
- *Archives of Computational Methods in Engineering*
- *Earth and Space Science*
- *Journal of Sports Analytics*
- *IISE Transactions*

Memberships

- American Statistical Association (ASA)
- Society for Industrial and Applied Mathematics (SIAM)
- Institution for Operations Research and Management Sciences (INFORMS)